



Faizan Shahid

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Home: Kharian (Pakistan)

EDUCATION AND TRAINING

Bachelor of Science in Chemical Engineering

National University of Sciences and Technology [19/10/2020 – 04/06/2024]

City: Islamabad | Country: Pakistan | Final grade: 3.55/4.00 | Thesis: Designed and modeled a 400,000 TPY Crude Terephthalic Acid Purification Plant, achieving 99.99% purity and 4.2-year projected payback.

Higher Secondary School (Intermediate) Education

Army Public College [05/05/2017 – 27/08/2019]

City: Kharian | Country: Pakistan | Final grade: A1

PROJECTS

ETFE Polymer Lining for Stainless Steel Corrosion Mitigation in HF Fume Environments

- Identified HF vapor-induced localized corrosion in stainless steel tanks.
- Specified ETFE lining for headspace region, extending equipment life by an estimated 10 years.
- Eliminated frequent patch repairs, saving approx. USD 50,000 annually.

Management of Change: Steam Trap Installation

- Led Management of Change for installation of 20+ steam traps across plant.
- Conducted thermodynamic sizing and specification of trap types (thermodynamic, inverted bucket).
- Executed PSSR and HAZOP actions; ensured safe tie-in during shutdown.

Designing a plant for Purification of Crude Terephthalic Acid

- Led Final Year Project on designing a 400,000-tonne/year purification plant, applying process design, simulation, and optimization techniques.
- Designed PFDs and P&IDs for a 400,000 TPY CTA purification plant.
- Modeled crystallization and solvent recovery units in Aspen HYSYS/Plus.
- Estimated capital and operating costs; projected 4.2-year payback.

Design and Simulation of Methanol Production plant from CO2 Hydrogenation

- Modelled catalytic conversion of Carbon dioxide (CO2) from coal power plant flue gases into methanol (CH3OH) using Aspen Plus Software, integrating a chemical absorption based capture unit.
- Parameter optimization resulted in a reduction of 1.6 tonnes of CO2 per tonne of synthesized methanol.

Simulink based Temperature control system analysis

- Developed a closed control loop in Simulink for EDIBON temperature control unit in university lab.
- Tuned the P, PI, and PID controllers to analyse system response and compared control system trends across different phases to evaluate performance improvements.

CFD case studies (Multiphase Flow and Segregation Profiles in Fluidized Bed Reactors)

- Generated mesh for different systems and vessels using Gambit.
- Simulated gas-solid integrations and segregation profiles in fluidised bed reactors using Ansys Fluent.
- Performed post processing analysis on the models created.

WORK EXPERIENCE

 **Fatima Fertilizers Limited** – Multan, Pakistan

City: Multan | Country: Pakistan

Shift Engineer

[22/07/2025 – Current]

- Managed production operations at Calcium Ammonium Nitrate plant, ensuring compliance with safety, environmental, and quality standards.
- Led troubleshooting of process disruptions, minimizing downtime and improving operational stability.
- Produced daily performance reports and analyses used for process optimization decisions.
- Coordinated with maintenance, project, and planning teams to resolve bottlenecks and execute shutdown activities.

 **Fatima Fertilizers Limited** – Multan, Pakistan

City: Multan | Country: Pakistan

Graduate Trainee Engineer

[22/07/2024 – 21/07/2025]

- Completed structured rotations across major plant sections, gaining hands-on experience in process control, instrumentation, and equipment performance.
- Conducted energy, efficiency, and quality monitoring; proposed operating adjustments under senior supervision.
- Participated in PSSR reviews, safety audits, and compliance checks across multiple units.
- Assisted in minor optimization projects targeting pressure balance, utility loads, and process stability.

CERTIFICATIONS

[Coursera, 03/05/2024]

Machine Learning

[Amal Academy, 05/06/2024]

Career Prep Fellow

- Completed an intensive program on career skills, including resume writing, interview prep, and networking.
- Led a team to organize and present a nationwide webinar on stress management for 100+ students.
- Managed content, speakers, promotion, and delivered the keynote presentation.

SKILLS

Technical Skills

Python / Aspen HYSYS & Aspen Plus / Auto-Cad / Ansys Workbench / Matlab/Simulink / Ansys Fluent / Technical Report Writing / Microsoft Office / Risk Assessment / Permit to Work System / CFD (Computational Fluid Dynamics)

Soft Skills

Teamwork / Adaptability / Conflict Resolution / Decision-making / Task Prioritization

VOLUNTEERING

Blood Donation drive

- Contributed over 40 hours during a one-month period to organize and lead the company-wide Blood Donation Drive in partnership with the MASH, Multan.
- Coordinated all operational needs, including securing the venue, managing inventory of supplies, and scheduling a team of 15 volunteers for donor registration and post-donation care.

Corporate CSR: Winter Clothing Drive Lead

- Pioneered and co-led the annual Corporate Social Responsibility (CSR) winter clothing collection drive across plant site units over a four-week period.
- Managed the setup and inventory tracking for 4 designated collection points and organized the sorting of over **45 bags** of winter apparel with a volunteer team.
- Coordinated the final delivery of all collected garments to the **Edhi Home**, providing essential winter wear for the residents.

Orphanage Charity & Supply Drive Coordinator

- Executed a digital and on-campus campaign to solicit donations, resulting in the successful collection of **PKR 0.2 Million**.
- Facilitated the safe and organized delivery of all collected resources directly to the orphanage, directly benefiting **60 children** residing at the facility.

National Literary Festival - Treasurer

- Directed the financial operations and reconciliation for a national-level university Literary Festival, managing the complete event budget with zero discrepancies.

Corporate Communications & Creative Development

- Developed and produced all professional graphic assets, including large-format visuals and digital presentations, for the Annual Corporate Awards Ceremony recognizing organizational achievement.
- Authored the original script for a stage play performed at the annual People Connect corporate event, successfully conveying key company values to a large employee audience.

LANGUAGE SKILLS

Mother tongue(s): Urdu

Other language(s): Pashto | English

HOBBIES AND INTERESTS

Football | Cricket | Gym | Tourism

ACHIEVEMENTS AND HONOURS

Academic

- Consistently achieved academic excellence, earning a place on the Dean's List twice in my undergraduate studies.
- Received the regional "Shaan e Aziz" Award for academic excellence in college board exams.
- Received the High Achievers Award for securing 3rd position in school board exams.

Non Academic

- Winner of the Ramzan Football Championship 2025 held at my workplace.
- Champion of the **NUST Inter-department Cricket** tournament with SCME Cricket Team.